Clinical Factors Associated with Disease Progression in Diabetes Patients

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Abstract

Background

Diabetes Mellitus is a chronic metabolic disorder characterized by elevated blood glucose levels and long-term complications affecting multiple organ systems. According to the World Health Organization (WHO), diabetes represents a major global health burden worldwide.

Objective

The objective of this project was to identify clinical variables associated with disease progression using data analysis and machine learning approaches.

Methods

A publicly available diabetes dataset obtained from the Scikit-learn library was analyzed using exploratory data analysis, correlation analysis, Linear Regression, and Random Forest Regression models.

Results

BMI demonstrated one of the strongest associations with disease progression across multiple analytical approaches. Linear Regression slightly outperformed Random Forest Regression in this dataset.

Conclusion

The findings support the use of clinical data analysis and predictive modelling for investigating factors associated with diabetes disease progression.

1. Introduction

Diabetes Mellitus is one of the leading chronic diseases worldwide and is associated with significant morbidity and mortality (WHO, 2025). Disease progression varies among individuals and may be influenced by multiple physiological and metabolic factors.

Understanding relationships between clinical variables and disease progression may support earlier intervention and improved patient management.

Objective of the Study

To analyze clinical variables associated with diabetes disease progression using exploratory data analysis and machine learning techniques.

2. Methods

2.1 Data Source

The dataset used in this analysis was obtained from the Scikit-learn machine learning library. The dataset contains standardized clinical measurements collected from diabetes patients.

2.2 Variables

Independent Variables

- Age
- Sex
- Body Mass Index (BMI)
- Blood Pressure
- Serum Measurements

Dependent Variable

- Quantitative measure of disease progression

2.3 Software and Tools

The following tools were used:

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

2.4 Dataset Preparation

The dataset used in this project was pre-standardized and suitable for analysis; therefore, no additional preprocessing steps were required.

Initial dataset inspection and exploratory data analysis were conducted prior to model development.

As all variables were standardized within the original dataset, some features contain negative values, representing measurements below the dataset mean rather than negative clinical measurements.

2.5 Data Analysis Workflow

The analysis workflow included:

1. Data loading and dataset inspection
2. Exploratory data analysis (EDA)

3. Correlation analysis
4. Predictive modeling using Linear Regression
5. Predictive modeling using Random Forest Regression
6. Model evaluation and comparison using R^2 score and Mean Absolute Error (MAE)

3. Code Availability

https://colab.research.google.com/drive/1I9QlVtWQzIWbf6QIvJGvzUg6mt9ZZixb#scrollTo=GMaguQzuz-H_

4. Results

4.1 Exploratory Data Analysis (EDA)

Research Questions

The exploratory data analysis focused on the following clinical questions:

1. Which clinical variables are most strongly associated with disease progression?
2. What is the relationship between BMI and disease progression?
3. What is the distribution of disease progression scores, and are there any potential outliers?

4.2 Correlation Analysis

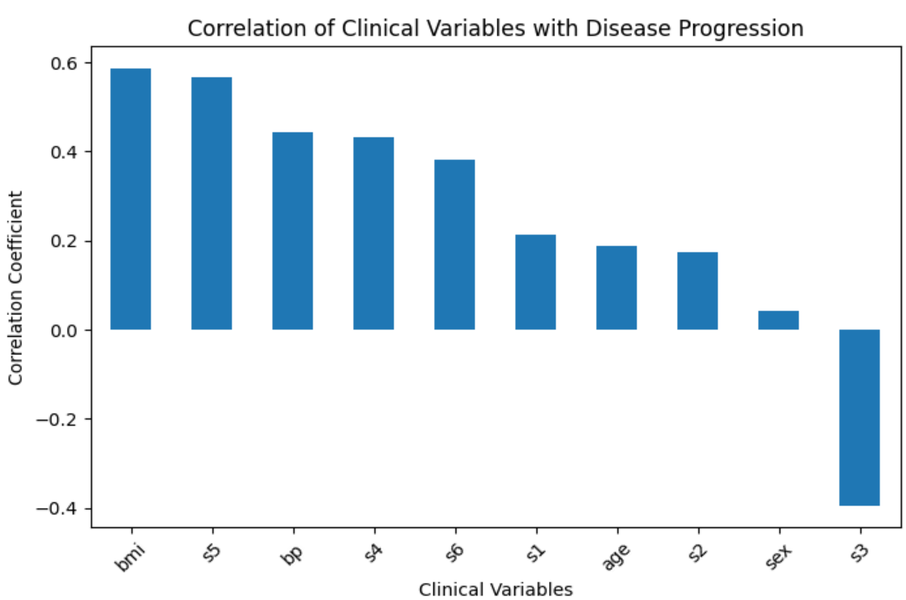
Code

Correlation analysis was performed to evaluate associations between clinical variables and disease progression.

Key Findings

- BMI demonstrated the strongest positive correlation with disease progression (r = 0.586)
- The variable s5 also demonstrated strong positive correlation (r = 0.566)
- Blood pressure showed moderate positive association (r = 0.441)
- The variable s3 demonstrated moderate negative correlation (r = -0.395)

Figure 1. Correlation of Clinical Variables with Disease Progression

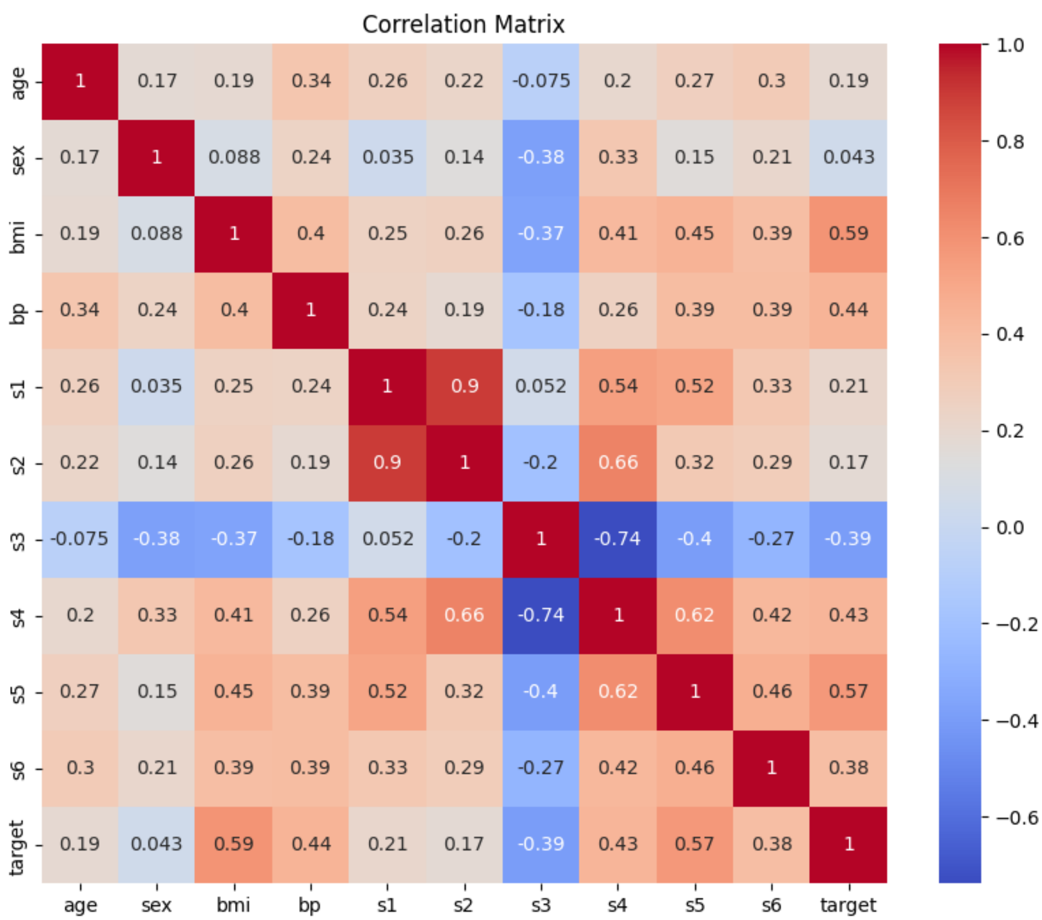


Interpretation

The results suggest that increased BMI may be associated with worsening disease progression in diabetes patients.

4.3 Correlation Matrix Analysis

Figure 2. Correlation Matrix of Clinical Variables



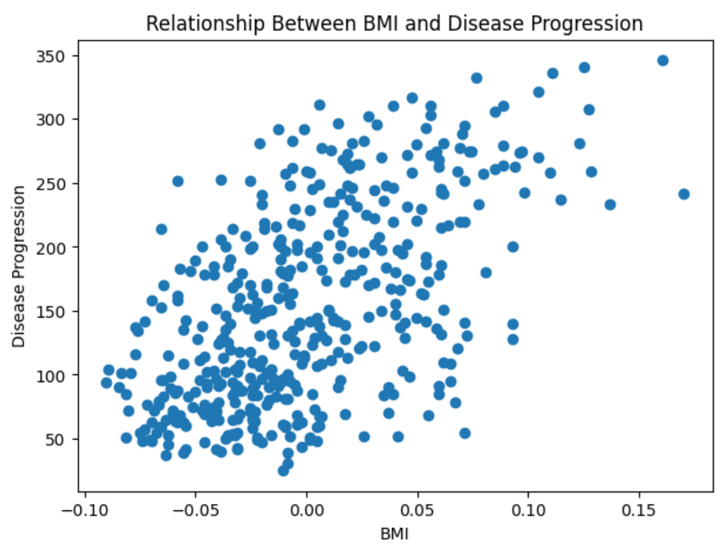
The analysis demonstrated that BMI and s5 showed the strongest positive relationships with the target variable.

The findings obtained from the correlation matrix were generally consistent with both the Random Forest and Linear Regression analyses.

4.4 Relationship Between BMI and Disease Progression

Scatter plot analysis demonstrated a positive relationship between BMI and disease progression.

Figure 3. Relationship Between BMI and Disease Progression



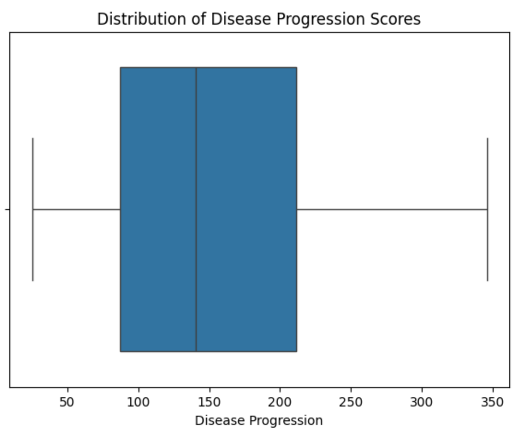
Higher BMI values were generally associated with increased disease progression scores.

Although biological systems are inherently complex and often non-linear, the observed relationship indicates that BMI may represent an important contributor to metabolic deterioration.

4.5 Distribution and Outlier Analysis

A boxplot was generated to evaluate the distribution of disease progression scores and identify potential outliers.

Figure 4. Distribution of Disease Progression Scores



The analysis demonstrated moderate variability in disease progression scores across patients.

The absence of extreme outliers suggests that the dataset was relatively stable and suitable for exploratory clinical analysis.

4.6 Machine Learning Model Development

Two predictive machine learning models were developed and compared:

1. Linear Regression

2. Random Forest Regression

4.7 Linear Regression Model

Linear Regression was selected as a baseline statistical model due to its interpretability and ability to model linear relationships between clinical variables and disease progression.

Linear Regression Results

Metric	**Value**
R^2 Score	0.453
Mean Absolute Error (MAE)	42.794

Coefficient Analysis

Feature	**Coefficient**
s5	736.20
bmi	542.43
s2	518.06
bp	347.70

Interpretation

The Linear Regression model demonstrated that BMI and s5 were among the strongest positive contributors to disease progression prediction.

The model achieved moderate predictive performance, suggesting that several relationships within the dataset can be reasonably approximated using linear modeling approaches.

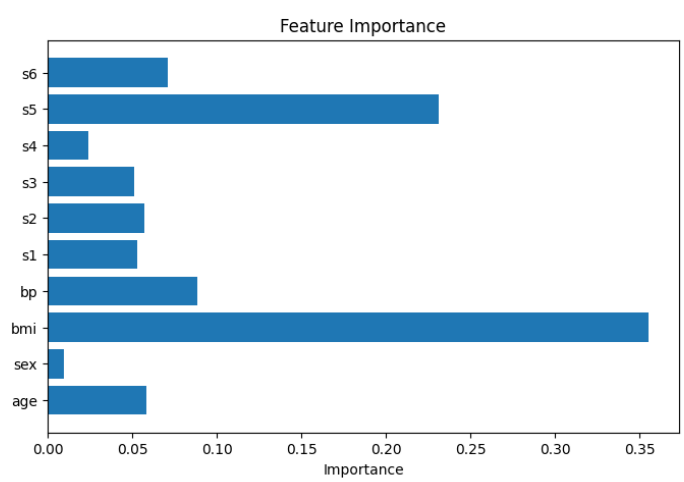
4.8 Random Forest Regression Model

Random Forest Regression was selected due to its ability to capture complex and potentially non-linear relationships between clinical variables and disease progression.

Random Forest Results

Metric	**Value**
R^2 Score	0.443
Mean Absolute Error (MAE)	44.053

Figure 5. Random Forest Feature Importance



Interpretation

The Random Forest model identified BMI and s5 as the most influential variables contributing to disease progression prediction.

Although Random Forest models are capable of capturing complex non-linear relationships, the model demonstrated slightly lower predictive performance compared to Linear Regression in this dataset.

4.9 Model Comparison

Both models were evaluated using:

- R^2 Score
- Mean Absolute Error (MAE)

Table 3. Comparison Between Predictive Models

Model	R^2 Score	MAE
Linear Regression	0.453	42.794
Random Forest Regression	0.443	44.053

Comparative Analysis

The Linear Regression model slightly outperformed the Random Forest Regression model across both evaluation metrics.

Although diabetes progression represents a biologically complex and potentially non-linear process, the available variables demonstrated relationships that could be reasonably approximated using linear modeling.

The comparison between statistical and machine learning approaches highlights the importance of selecting models based on dataset characteristics rather than model complexity alone.

5. Discussion

The analysis demonstrated that BMI was significantly associated with disease progression in patients with Diabetes Mellitus.

These findings are consistent with existing clinical knowledge linking obesity and metabolic dysfunction to worsening disease outcomes.

Clinical Implications

- BMI may serve as an important indicator for risk assessment
- Monitoring metabolic parameters could support early intervention strategies
- Data-driven approaches may assist clinical decision-making

Limitations

- Limited dataset size
- Educational dataset rather than a large real-world clinical cohort

- Lack of longitudinal patient follow-up
- Biological systems are inherently complex and may involve non-linear interactions not fully captured within the available variables

Future Directions

Future analyses could include:

- Larger patient populations
- More diverse clinical variables
- Longitudinal patient measurements
- Advanced predictive modeling approaches

6. Conclusion

This project demonstrates the integration of clinical understanding, exploratory data analysis, machine learning, and professional medical writing.

The analysis identified BMI as one of the strongest predictors of diabetes disease progression across multiple analytical approaches, including correlation analysis, Linear Regression, and Random Forest Regression.

Comparison between predictive models demonstrated that the Linear Regression model slightly outperformed the Random Forest model in this dataset.

This portfolio project demonstrates the ability to:

- Perform exploratory clinical data analysis
- Apply statistical and machine learning models
- Interpret biomedical findings
- Evaluate and compare predictive models
- Translate analytical results into structured scientific documentation

7. References

1. World Health Organization (WHO) – Diabetes Fact Sheet
2. American Diabetes Association (ADA) – Standards of Care in Diabetes
3. Scikit-learn – Diabetes Dataset Documentation
4. Scikit-learn – Linear Regression Documentation
5. Scikit-learn – Random Forest Regression Documentation